

- Decision Theory

↳ Decision theory is the science of making optimal decisions in the face of uncertainty.

↳ Statistical decision theory is concerned with the making of decisions when in the presence of statistical knowledge (data) which shed light on some of uncertainties involved in decision problem.

↳ It is a systematic method for evaluating decision rules in ambiguous circumstances.

↳ It matches the traditional issue of estimation and hypothesis testing in statistics.

↳ Statistical decision theory is a formal setup to analyze decision rules in the context uncertainty.

↳ ~~The standard~~

Component of decisions theory

↳ Decision maker

↳ course of action

↳ state of nature

↳ ~~Sta~~ Pay-off.

↳ Pay-off table

↳

Procedure of Decision

- ↳ All possibilities are identified.
- ↳ Then compute the cost/profit
- ↳ Take action

Type of Decision making

- ↳ Under certainty → complete info.
 - ↳ Under risk → state of nature is unknown (Probability)
 - ↳ Under uncertainty → Hardly any knowledge about state of nature (no Prob.)
 - ↳ Under conflict → Rational considering
- take action, counter action
 - ↳ Under partial information →
- Statistical Decision theory is a systematic method for evaluating decision rules in ambiguous circumstances.
- It is framework for inference for any formally defined decision-making problem.

Ex Drug company is deciding whether or not to sell a new pain reliever.

Here decision making processes usually involve uncertainty. The main goal of this study (ex!) is to find an action or decision rule that reduces our loss as much as possible.

Three steps

- ↳ loss function
- ↳ decision rule.
- ↳ action